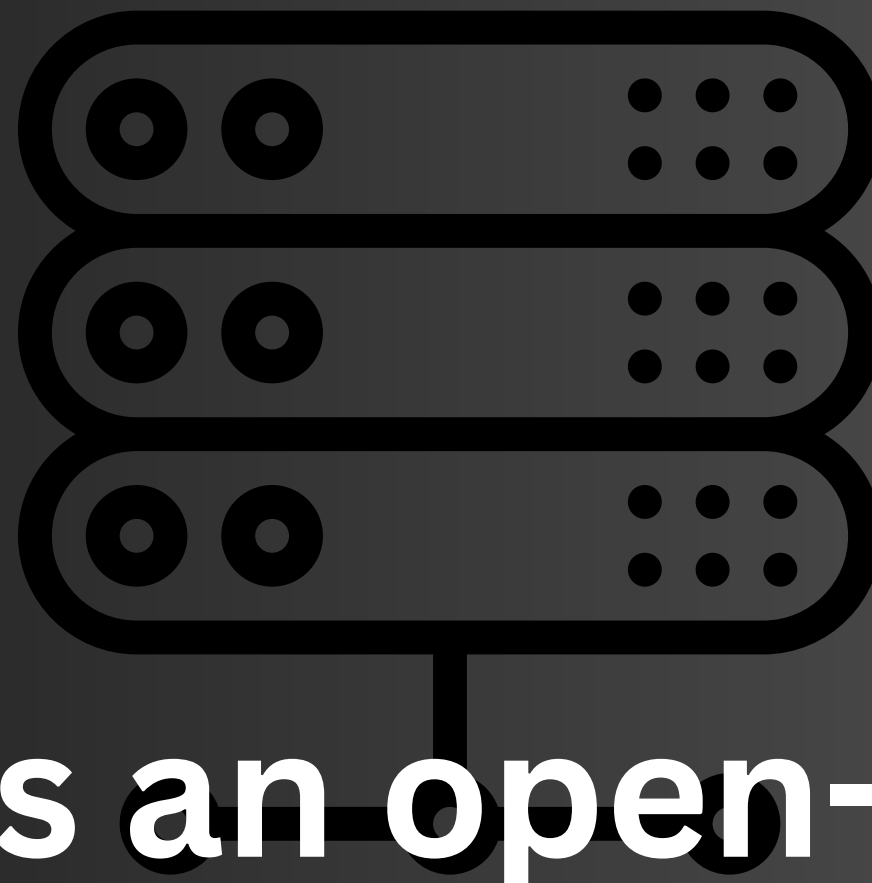




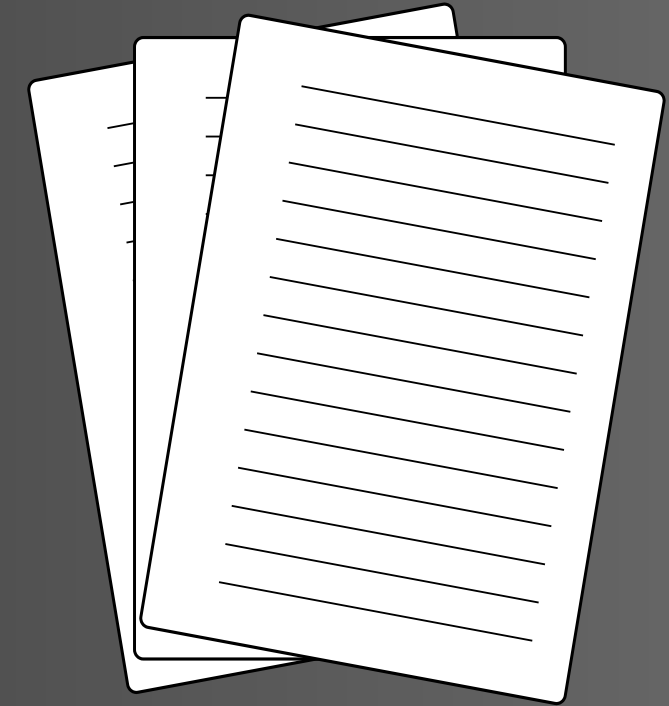
What is Terraform?





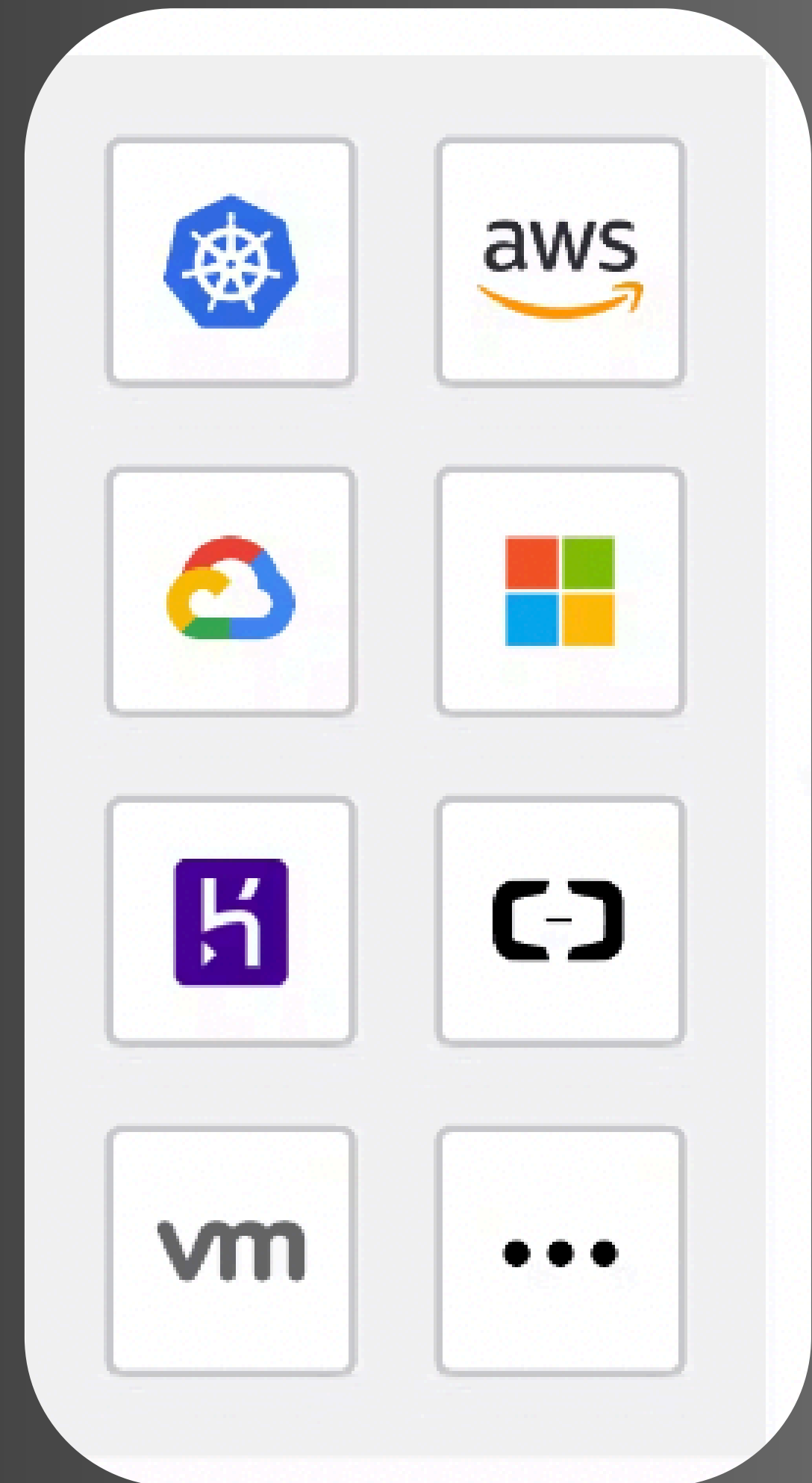
**Terraform is an open-source
infrastructure as code (IaC) tool.**

What is IaC?



It enables you to define, provision, and manage infrastructure across multiple cloud providers using a **declarative configuration language**.

It supports multi-cloud environments, allowing consistent and efficient infrastructure management.





AWS EC2 Instance

Which cloud provider?

Which location?

What type of resource?

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "5.54.1"
    }
  }
}

provider "aws" {
  region = var.region
}

resource "aws_instance" "myserver" {
  ami           = "ami-0c0e147c706360bd7"
  instance_type = "t3.nano"

  tags = {
    Name = "SampleServer"
  }
}
```

Terraform Config

- It uses **.tf** extension
- Format is HCL (Hashicorp Config Language)
- Declarative Language

Terraform supports JSON format also

- HCL:

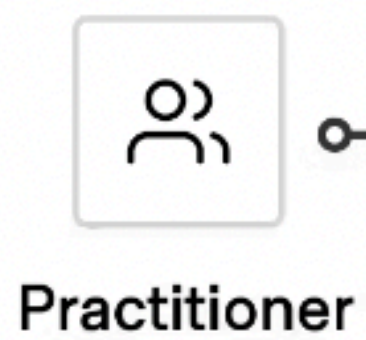
hcl

```
provider "aws" {  
  region = "us-east-1"  
}
```

- JSON:

json

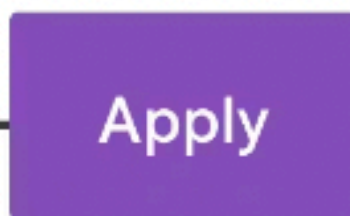
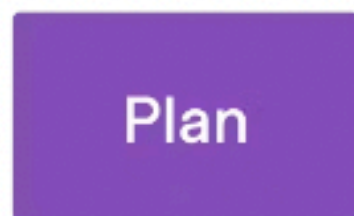
```
{  
  "provider": {  
    "aws": {  
      "region": "us-east-1"  
    }  
  }  
}
```

Practitioner



Infrastructure
as Code



- **terraform init**
- **terraform plan**
- **terraform apply/destroy**

- "**Consistency**: Define infrastructure in code"
- "**Automation**: Reduce manual work"
- "**Repeatable**: Replicate environments easily"

State Management

The state file (**terraform.tfstate**) maintains a detailed record of the current state of managed resources

This state file can be stored locally or remotely, with remote storage options enabling collaboration by sharing the state across teams and environments.

Why Terraform?

- **Automate Setup:** Quickly create and configure resources like servers, databases, and networks.
- **Consistency:** Ensure every environment is set up the same way, reducing errors.
- **Scalability:** Easily replicate and scale infrastructure for different environments.
- **Version Control:** Track and manage infrastructure changes just like software code.
- **Flexibility:** Works with multiple cloud providers and on-premises setups.

Multiple Resources using

- Count
- for_each

Terraform Modules

Terraform Modules:

Modules are containers for multiple resources that are used together.

A module consists of a collection of **.tf and/or .tf.json** files kept together in a directory.

Modules are the main way to package and reuse resource configurations with Terraform.

Workspaces

tfstate

tfstate

tfstate

workspace-dev

workspace-test

workspace-prod



tf config

Terraform Cloud

Terraform Cloud is a managed service provided by HashiCorp that facilitates collaboration on Terraform configurations.

Providing features like

- **remote state management,**
- **version control system (VCS) integration,**
- **automated runs, and**
- **secure variable management.**



Region

Availability Zone

Availability Zone



VPC



Public subnet



Public subnet



Security group



Web servers



Web servers



Private subnet



Private subnet



S3 gateway

Security group



Database servers



Database servers

